

Cloud Kubernetes vs On-Prem Kubernetes

Cloud	On-Prem
Managed Control Plane	Self Managed
Elastic Infrastructure	Fixed Hardware
Internet Available	Often Private
Easy Scaling	Hardware Dependent
Pay-as-you-go	Capital Investment

Discussion

When should we choose each?

Why Hybrid Wins

Organizations rarely choose only cloud.

Reality

AWS Azure VMware Bare Metal Edge IoT Factories

Need one Kubernetes platform.

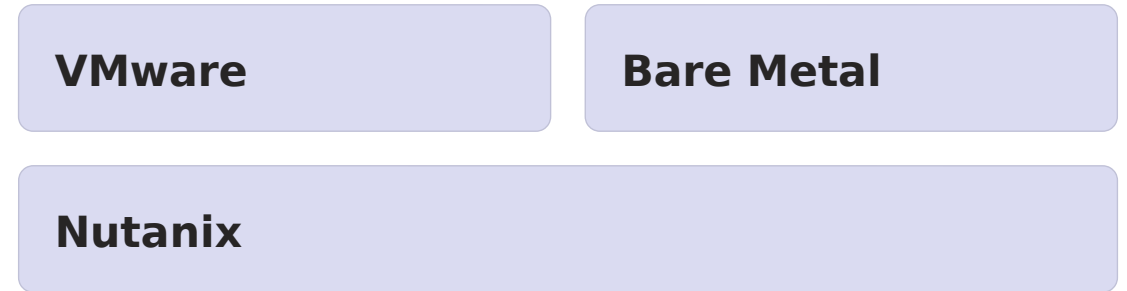
Hybrid enables this.

EKS Anywhere Architecture

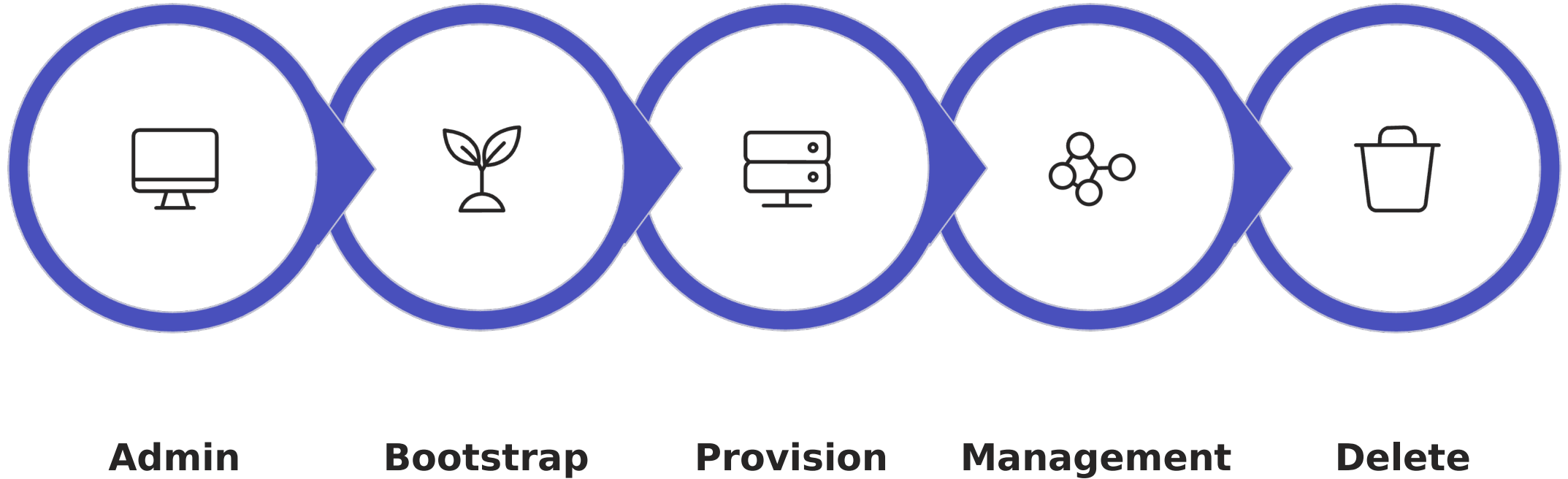
Architecture Components



Infrastructure

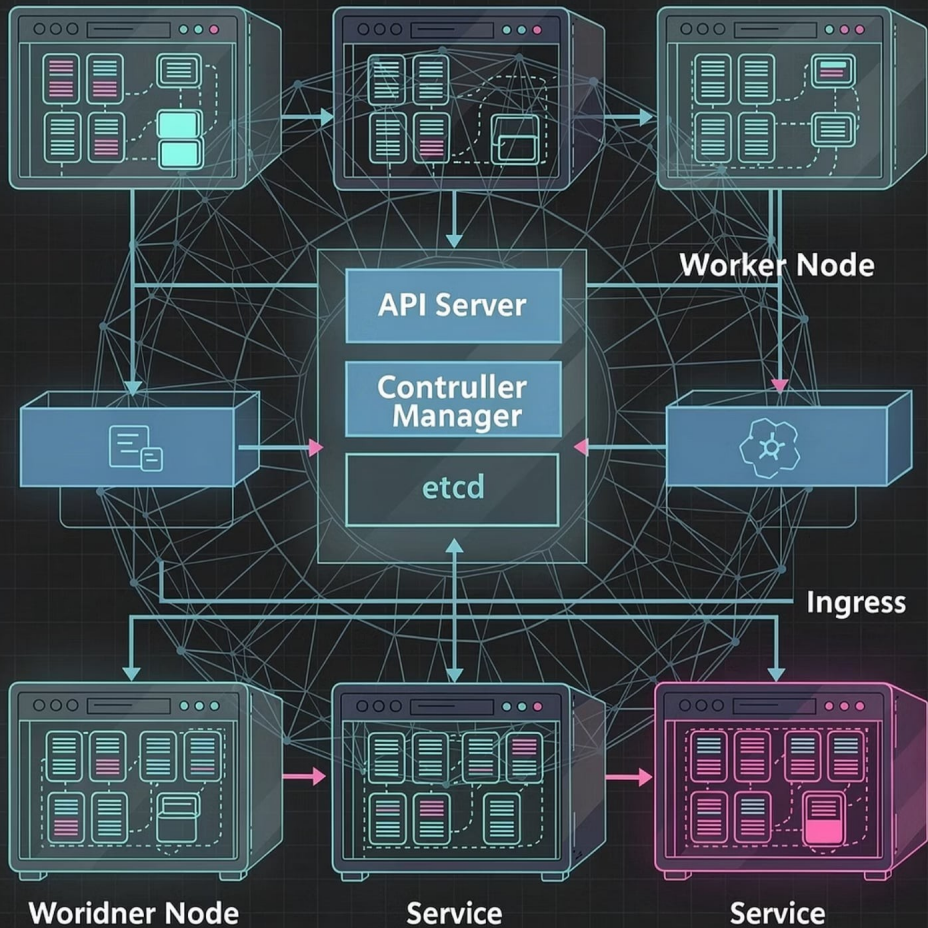


EKS Anywhere Deployment Flow



Management Cluster remains.

Control Plane



Cluster Components

Control Plane

- API Server
- Scheduler
- Controller Manager
- etcd

Worker Nodes

- kubelet
- kube-proxy
- Container Runtime

Applications

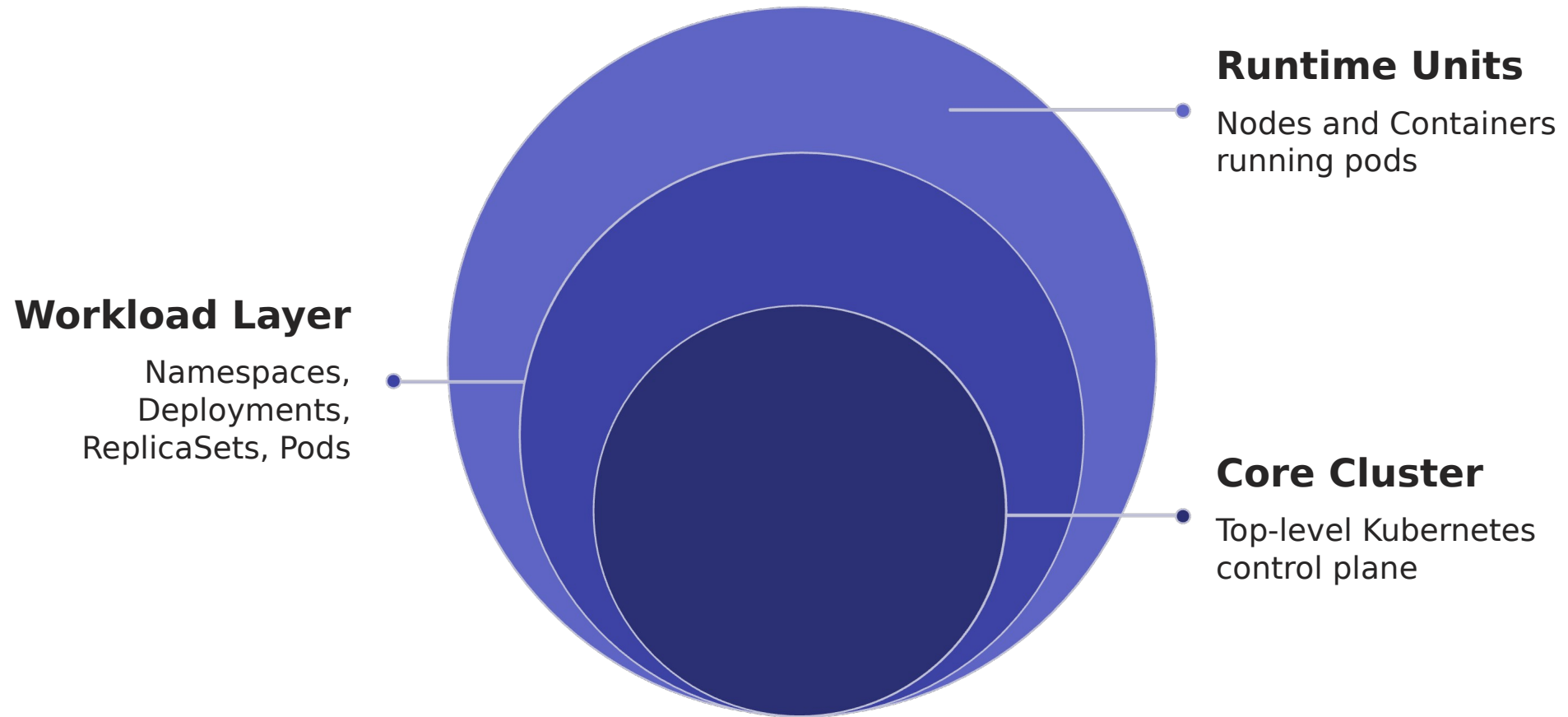
Pods

Services

Ingress

Volumes

Kubernetes Object Hierarchy



Understand hierarchy before labs.

Cluster API (CAPI) Overview

Cluster API

An open-source Kubernetes project

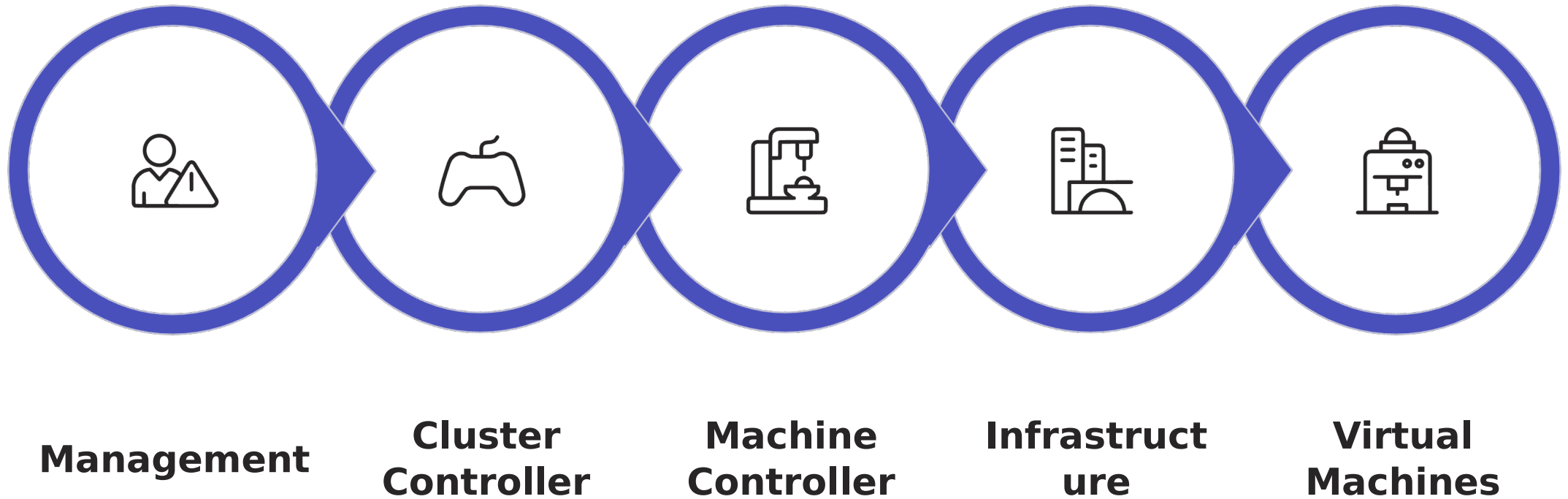
Purpose

Manage Kubernetes clusters using Kubernetes.

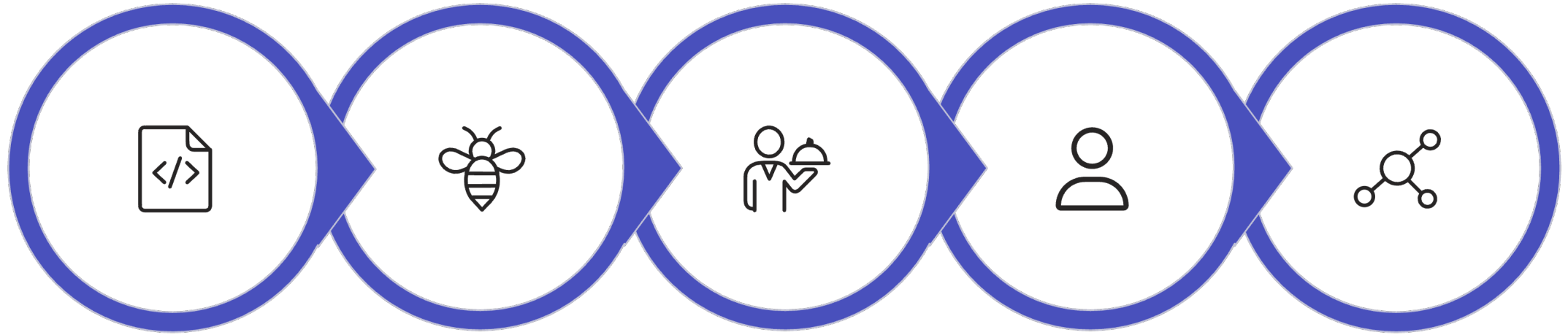
Everything becomes Kubernetes objects.

Infrastructure becomes declarative.

Cluster API Components



Cluster API Workflow



**Cluster
YAML**

Cluster API

Create VMs

Install K8s

Join Nodes

Infrastructure becomes code.

Summary

Today We Learned

01

Enterprise Kubernetes

02

Enterprise Challenges

03

EKS Anywhere

04

Platform Engineering

05

Hybrid Kubernetes

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Cloud vs On-Prem

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EKS Anywhere Architecture

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Cluster Components

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Cluster API Basics